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MILITARY SPECIFICATION

BOILERS, STEAM AND HOT WATER, WATERTUBE (COIL AND SERPENTINE), PACKAGED TYPE (40,000 TO 10,000,000 BTU/HR THERMAL OUTPUT CAPACITY)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers watertube (coil and serpentine) packaged steam boiler or hot water generator units.

1.2 Classification. The steam boiler or hot water generator shall be of the following types, sizes, and classes as specified (see 6.2 and 6.5):

TYPES

Type I - Steam boiler producing low pressure steam not greater than 15 pounds per square inch gage pressure (psig).
Type II - Steam boiler producing high pressure steam between 16 and 150 psig.
Type III - Steam boiler producing high pressure steam between 151 and 350 psig.
Type IV - Steam boiler producing high pressure steam between 351 and 600 psig.
Type V - Hot water generator producing low temperature water not greater than 160 psig or 250 degrees Fahrenheit (^{o}F).

|Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.|

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SIZES

Size 1 - Thermal output capacity between 40,000 and 320,000 British thermal units per hour (Btu/hr).
Size 2 - Thermal output capacity between 320,001 and 10,000,000 Btu/hr.

CLASSES

Class 2 - Light oil fired.
Class 3 - Gas fired.
Class 5 - Combination light oil and gas fired.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications. The following specifications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-B-3180 - Boilers and Related Equipment, Packaging of.
MIL-B-18796 - Burners, Single: Oil, Gas, and Gas-Oil Combination for Packaged Boilers (320,000 to 125,000,000 Btu/hr Thermal Output Capacity).
MIL-B-18797 - Burners, Single: Light Oil, Gas, and Light Oil-Gas Combination for Packaged Boilers (Up to 320,000 Btu/hr Thermal Output Capacity).

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents which are current on the date of the solicitation (see 6.2).

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AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME Boiler and Pressure Vessel Code.
Section I - Power Boilers.
Section IV - Heating Boilers.
Section VIII,
Division I - Pressure Vessel.
Section IX - Welding and Brazing Qualifications.
CSD-1 - Controls and Safety Devices for Automatically Fired Boilers.
Performance Test Code (PTC).
PTC 4.1 - Steam Generating Units.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

HYDRONICS INSTITUTE

Testing and Rating Standard for Heating Boilers.

(Application for copies should be addressed to the Hydronics Institute, 35 Russo Place, P.O. Box 218, Berkeley Heights, New Jersey, 07922.)

UNDERWRITERS LABORATORIES INC. (UL)

UL-726 - Oil-Fired Boiler Assemblies.
UL-795 - Commercial-Industrial Gas-Heating Equipment.

(Application for copies should be addressed to the Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The packaged unit shall either be a steam boiler or a hot water generator as specified (see 6.2). The steam boiler or hot water generator shall be of a packaged serpentine or coil watertube design. Serpentine watertube boilers shall not be used for type IV service. The steam boiler or hot water generator shall be capable of producing the thermal output capacity specified (see 6.2). The steam boiler shall be furnished complete with all appurtenances specified herein and shall be factory wired and ready for operation when connected to water, fuel, and electrical supplies. The steam

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boiler or hot water generator shall be fired on the fuel specified (see 6.1.1 and 6.2). The steam boiler or hot water generator shall be furnished complete with the burner specified in MIL-B-18796 or MIL-B-18797 as applicable.

3.2 Codes and standards. The steam boiler or hot water generator shall conform to the applicable requirements of UL-726, UL-795, Section I, Section IV, Section VIII, Division I and Section IX of the ASME Boiler and Pressure Vessel Code (hereinafter referred to as the ASME Code), ASME CSD-1 and PTC 4.1.

3.3 Standard commercial product. The steam boiler or hot water generator shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the ### being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.3.1 System of measurement. The dimensions used in this specification are not intended to preclude the use of the metric system of measurement in the fabrication and production of the material, individual parts, and the finished product, provided form, fit, and function requirements are satisfied.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification.

3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to ensure interchangeability of component parts, assemblies, accessories, and spare parts.

3.6 Design. The steam boiler or hot water generator shall be designed to withstand the normal strains, jars, and vibrations incidental to shipping, storage, and installation. The steam boiler or hot water generator shall be designed for an indoor or outdoor installation as specified (see 6.2). The steam boiler or hot water generator shall be designed for the following operating parameters as specified (see 6.2):

a. Design pressure (psig).
b. Operating pressure at packaged unit outlet (psig).
c. Design temperature (^{o}F).

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d. Operating temperature at packaged unit outlet (°F).
e. Maximum continuous thermal output (Btu/hr).
f. Feedwater temperature (°F).
g. Maximum, minimum, and average anticipated ambient air temperatures (°F).
h. Site elevation above sea level (feet).

3.7 Performance characteristics.

3.7.1 Steam purity. When a type II, type III, or type IV steam boiler is required (see 3.1), the moisture content of the steam leaving the steam separator shall be not greater than 1.5 percent.

3.7.2 Thermal efficiency. When tested in accordance with 4.5.2, the thermal efficiency shall be not less than 81 percent for light oil fuels and not less than 81 percent for gas fuel, except for type I, size 1 steam boilers or type V, size 1 hot water generators. Type I, size 1 steam boilers or type V, size 1 hot water generators shall have a thermal efficiency of not less than 75 percent at any firing rate.

3.7.3 Heat input rate. When the steam boiler or hot water generator is operating at maximum output, the heat input rate shall be not greater than 25,000 Btu/hr per square foot of effective heating surface (see 6.5.1).

3.7.4 Exit temperature at stack. Unless otherwise specified (see 6.2), the exit temperature of the flue gas at the stack shall be not less than 325°F without an economizer. When exit temperatures of less than 325°F without an economizer are allowed, the flue gas passages shall be of corrosion resistant materials to prevent potential flue gas condensate corrosion problems.

When operating at minimum load and an economizer is required (see 6.2), the exiting flue gas temperature shall be not less than 300°F. The economizer water temperature at flue gas exit shall not be less than 220°F. When exit flue gas temperatures of less than 300°F are allowed use corrosion resistant materials in the flue gas passages and economizer to prevent potential flue gas condensate corrosion problems.

3.7.5 Thermal shock. The steam boiler or hot water generator shall be designed to withstand the thermal stresses associated with cold start and emergency shutdown operating conditions.

3.8 Environmental requirements. The emission requirements shall be met at the maximum required continuous output. The steam boiler or hot water generator and burner shall meet the legal emission requirements required by the local, state, and Federal environmental rules and regulations. The emission requirements for opacity, particulates, nitrogen oxides, carbon monoxide, and sulfur dioxide shall be as specified (see 6.1.1 and 6.2).

3.9 Construction. Steam boilers and hot water generators shall be constructed for the specified design conditions and performance requirements in accordance with the applicable sections of the ASME Code, UL-726, UL-795 and the additional requirements specified herein.

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3.9.1 Hot water generator. When furnished (see 3.1), a hot water generator shall be designed to operate in a flooded condition without a steam space for pressure control. The expansion or pressure regulating vessel, feed water regulating valve, and associated controls are not considered part of this specification when a hot water generator is required (see 6.1.1).

3.9.2 Water pressure drop. Unless otherwise specified (see 6.2), when a hot water generator is furnished (see 3.1), the water pressure drop shall be not greater than 10 psig. Flow restrictions causing temporary pressure drops shall not cause cavitation or flashing of the hot water when the generator is operated at the required temperature and pressure.

3.9.3 Furnace. Air or water wall headers and supply tubes shall be protected from the products of combustion. The walls and floor of the furnace shall permit free expansion of the steam boiler or hot water generator without undue thermal stress on any part of the steam boiler or hot water generator. Refractory lined fireboxes shall be lined with a performed, castable, or plastic refractory. Refractory brick shall not be acceptable.

3.9.4 Refractories. Refractories shall withstand the temperature existing under maximum load conditions. The refractories shall be formed in place or cast in sections of such size as to be easily replaceable through existing openings. The refractory sections shall be designed and secured in position in such a manner as to withstand the vibration and shocks occurring during shipment.

3.9.5 Access door and observation ports. The steam boiler or hot water generator shall be equipped with an access door for normal field internal inspection, cleaning and maintenance of the furnace fireside. The steam boiler or hot water generator shall be equipped with cover plates or handholes to allow for internal inspection and cleaning. Not less than one observation port shall be provided to permit operators to see the furnace interior and flame pattern.

3.9.6 Tubes. Tubes shall either be electric-resistance welded or seamless steel. Spinners, turbulators, and other such devices shall not be permitted in the tubes.

3.9.7 Drainage. The circulation system shall be equipped with a valve(s) to permit complete drainage by gravity or by blowing with compressed air of the tubes and vessels.

3.9.8 Insulation. Insulation shall be in accordance with the requirements of UL-726 or UL-795. Asbestos shall not be used for gaskets or insulation materials.

3.9.9 Inner casing. When an inner casing is included in the manufacturer's standard design, it shall be pressure-tight to prevent the products of combustion from coming in contact with the steam boiler or hot water generator insulation and outer casing. The inner casing may either be the tube wall itself or a separate inner casing having a thickness of not less than 0.135 inch, placed just behind the tubes or the refractory backing.

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3.10 Steam boiler and hot water generator trim. Each steam boiler or hot water generator shall be fully equipped with the trim required under the applicable section of the ASME Code for the type of unit specified herein. The additional appurtenances specified herein shall also be furnished. When disassembly is required for shipping clearance and prevention of damage during shipment, such trim and appurtenances may be field installed. The trim, valves, connecting piping and fittings shall be suitable for the design pressure and temperature as specified herein. Instrumentation, limit controls, and interlock switches shall be in accordance with the applicable requirements of ASME CSD-1. Signals from switches and alarms shall be compatible with the control system required by MIL-B-18796 or MIL-B-18797 as applicable. Unless otherwise specified (see 6.2), the trim and appurtenances shall include but not necessarily be limited to the following:

a. High water level alarm for steam boilers.
b. Low water level alarm and a low water level fuel cutoff switch. The switch shall be mounted on a separate bridle with a drain valve.
c. A manual bypass momentary contact push button switch for the low water level cutoff switch of item b shall be located adjacent to the drain valves for the low water cutoff switch.
d. Piping and valves on steam boilers for continuous blowdown. The continuous blowdown valves shall have stainless steel trim and be able to maintain the position of a manual setting.
e. A top drum vent valve with threaded connection and corrosion resistant trim.
f. A chemical feed pipe connection with a diameter of not less than 0.75 inch with a corrosion resistant trimmed stop valve installed on the feedwater inlet piping.

3.11 Feedwater regulator. Steam boilers, when furnished, shall be provided with an automatic feedwater regulating system to maintain the boiler manufacturer's established steaming water level (see 6.1.1). The regulating system shall maintain the setpoint within $\pm$5 percent of the full range of allowable operating level. Feedwater regulators shall include the required sensing, actuating, regulating, and controlling equipment necessary to perform under the required conditions specified herein. The regulating system shall be provided complete with all necessary fittings, tubing, piping, mechanical linkages, wiring and other material or parts required to connect the operating components.

3.11.1 Feedwater regulating valve. Regulating valves shall be furnished with a double block and bleed piping bypass to allow the regulator valve to be removed while the boiler is operating. The regulating valve shall be equipped with a handwheel or knob for manual control and operation and shall have the manufacturer's standard position indicator. The valve shall be constructed with a balanced force design to allow the valve to move easily against the force of the feedwater flow. Materials of construction shall be the manufacturer's standard for good corrosion resistance in boiler feedwater service.

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3.11.2 Manual feedwater regulator control. When specified (see 6.2), the feedwater regulator system shall be furnished with a manual and automatic switch and a control mechanism for manually adjusting the feedwater regulator position. The direction in which the control mechanism should be operated to raise or lower the water level shall be clearly indicated.

3.12 Burners. Size 1 steam boilers or hot water generators shall be furnished with a factory installed burner, ignition system, programmed control system, and associated fuel system in accordance with MIL-B-18797. Size 2 steam boilers or hot water generators shall be furnished with a factory installed burner, ignition system, programmed control system, and associated fuel system in accordance with MIL-B-18796.

3.12.1 Burner combustion control system. The burner firing rate shall be controlled as a function of the outlet pressure on steam boilers or outlet temperature on hot water generators. The burner combustion control system shall be in accordance with MIL-B-18796 or MIL-B-18797 as applicable. When specified (see 6.1.1 and 6.2), the combustion control system shall be furnished with an oxygen compensation system and shall be one of the following as specified (see 6.2).

a. Fixed rate, on-off control.
b. Three position high-low-off control.
c. Single point positioning control.
d. Parallel position control.

3.12.2 Control panel. The controls for the steam boiler or hot water generator and burner, including operating switches, indicating lights, gages, alarms, motor starters, fuses, and control system circuits shall be mounted on a single control panel or cabinet insofar as practicable in order to centralize the control functions. The location of the panel shall either be at the side or back of the steam boiler or hot water generator, or in a free-standing control panel for remote mounting as specified (see 6.2). When a free-standing remote mounted control panel is to be furnished, skid wiring or tubing shall terminate in a skid mounted junction box.

3.13 Electrical requirements. The steam boiler or hot water generator and associated equipment shall obtain power from a circuit with a single master manual service disconnect switch and automatic circuit breaker for all electrical equipment.

3.14 Cleaning, treatment, and painting. Bare steel surfaces shall be cleaned and a primer coat applied. Paint shall be applied to normally painted surfaces in good commercial practice.

3.15 Mounted operating instructions. Unless otherwise specified (see 6.2), operating instructions shall be summarized on a decal or corrosion resistant plate to be mounted inside the control panel door or similar weatherproofed location.

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3.16 Identification marking. Identification shall be permanently and legibly marked directly on the steam boiler or hot water generator or on a corrosion-resistant metal plate. Identification shall include the ASME Code stamp, manufacturer's name, unit model and serial number, thermal capacity, year of manufacture, maximum allowable working pressure and other pertinent information as identified in the ASME codes. The Department of Defense contract number that this equipment is procured under shall also be marked on the equipment or identification plate.

3.17 Spare parts. When specified (see 6.2), spare parts shall be furnished and shipped with each steam boiler or hot water generator. When furnished, the spare parts required and the quantity thereof shall be as specified (see 6.2).

3.18 Factory start-up service. Unless otherwise specified (see 6.1.1 and 6.2), factory authorized service personnel shall be provided to supervise installation and start-up of the packaged steam boiler or hot water generator for one 8-hour working day.

3.19 Workmanship. Workmanship shall be of such quality as to produce steam boilers or hot water generators in accordance with the ASME Code, as applicable, and shall meet the requirements of this specification and standards prevailing among manufacturer's who normally produce this type of equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements. Documentation of inspection shall be made available to the Government upon request.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this document shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Quality conformance inspection (see 4.2.1).
b. On-site inspection (see 4.2.2).

4.2.1 Quality conformance. This inspection shall be performed at the manufacturer's facilities. Unless otherwise specified (see 6.2), the quality conformance inspection shall include the examination of 4.3, the pressure test of 4.4, and the packaging inspection of 4.6.

4.2.2 On-site inspection. The on-site inspection shall be in addition to the quality conformance inspection performed at the manufacturer's facilities. Unless otherwise specified (see 6.2), the on-site inspection shall consist of the tests of 4.5. When specified by the contracting officer (see 6.2), all or part of the on-site inspection may be performed at the manufacturer's facilities. When specified (see 6.2), the examination of 4.3 and the test of 4.4 shall be repeated on-site.

4.3 Examination. Each boiler or hot water generator and related equipment supplied under this specification shall be examined for compliance with the requirements specified in section 3 of this document. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirement or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Fuel systems. The preassembled fuel system that is furnished integrally with the steam boiler or hot water generator shall be checked for leakage during the fire testing of the unit. The soap bubble or the halogen-sniff test method shall be used to verify integrity of the gas fuel system.

4.5 On-site tests. Operational fire tests and auxiliary equipment tests shall be performed on each installed steam boiler or hot water generator. These tests shall be in addition to any manufacturer's standard fire test procedure, unless compliance evidence provided by the manufacturer is accepted at the discretion of the contracting officer or authorized representative. Acceptable fire test results from identical or similar equipment installed at the same site may be used at the discretion of the contracting officer as proof that equipment procured with this specification is in compliance with one or more of the tests of 4.5.1 through 4.5.8.

4.5.1 On-site operational tests. The steam boiler or hot water generator shall be tested to demonstrate control and operational conformance to the requirements of this specification and MIL-B-18796, or MIL-B-18797, as applicable, under varying load conditions ranging from the specified maximum capacity to the minimum turndown ratio without on-off cycling. Automatic burners shall also be tested for ability to provide steam or hot water in accordance with demand when on-off cycling is required. A steam boiler or hot water generator that exhibits (1) excessive or unexplained loss of ignition,

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(2) nuisance shutdown due to faulty burner or control operation, (3) improper flame, (4) excessive carbon deposits, or (5) necessity for frequent or difficult adjustments shall be rejected.

4.5.2 Capacity and efficiency tests. The capacity and efficiency of type I, size 1 steam boilers or type V, size 1 hot water generators shall be determined in accordance with the Hydronics Institue Testing and Rating Standard for Heating Boilers. The capacity and efficiency and if required, steam quality at the maximum specified capacity of all other steam boilers or hot water generators shall be determined in accordance with the abbreviated efficiency test of ASME PTC 4.1 using the heat loss method. Performance shall be as specified in 3.7. Methods of measuring the flow of steam, water, and fuel other than weighing will be allowed. The accuracy of the method shall assure the Government the desired performance specified in 3.7 and be acceptable to the contracting officer or authorized representatives. Failure to meet the requirements of 3.7 shall constitute cause for rejection.

4.5.3 Equipment tests. Control panels, insulation, valves, controllers, regulators, electric motors, dampers, and other related equipment shall be operated as part of the capacity and efficiency tests. The operation of this equipment shall be closely observed during the operational tests for possible defects or nonconformance. The action of dampers, actuators, and valves shall be smooth without backlash.

4.5.4 Sequencing. The steam boiler or hot water generator shall start, operate, and stop in strict accordance with the specified operating sequence.

4.5.5 Burner tests. Burner tests shall be conducted on the burner as specified in MIL-B-18796 or MIL-B-18797 in addition to the testing specified herein.

4.5.6 Limit and cutoff switches. Safety shutdowns shall be caused by simulating fuel safety interlock actuating conditions for each steam boiler or hot water generator limit and cutoff switch specified in 3.10. The safety shutdowns shall occur without damage to the equipment.

4.5.7 Blowdown valves and try cocks. All blowdown valves and try cocks shall be tested for proper operation.

4.5.8 Emissions testing. The required emissions testing for opacity, particulates, nitrogen oxides, carbon monoxide, and sulfur dioxide as required in 3.8 shall be conducted in accordance with the procedures of the governing Federal, state, or local environmental agency as specified (see 6.2).

4.6 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of section 4 of MIL-B-3180.

5. PACKAGING

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5.1 Preservation, packing, and marking. The steam boiler or hot water generator shall be preserved, packed, and marked in accordance with MIL-B-3180. The level of preservation and level of packing shall be as specified (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The steam boilers and hot water generators covered by this specification range from 15 to 600 psig and are intended for generation of steam and hot water used in processing operations and space heating systems in land installations.

6.1.1 Selection factors. The following should be considered in the selection of a steam boiler or hot water generator and the exercise of options herein:

a. The fuel class should be determined in accordance with the availability or economics of the fuel supply and any DOD requirements for capability of firing multiple fuels (see 3.1).
b. If the steam boiler or hot water generator is subject to environmental regulations, each agency should consult with their environmental support office since regulation limits of particulate and emissions for they vary from city, state, and Federal agencies (see 3.8).
c. The expansion vessel, water circulating pump, feedwater regulator and associated controls are not part of this specification because this equipment has been considered beyond the normal scope of a packaged hot water generator unit (see 3.9.1).
d. Hot water generator systems should use a nitrogen pressured vessel for pressure control. The pressure control vessel is not part of the typical packaged unit. The feedwater controller for the hot water generator should be mounted on the nitrogen pressured vessel (see 3.11).
e. Oxygen compensation systems should not be specified for on-off or three position high-low-off combustion controls (see 3.12.1).
f. Factory service should be specified only if the boiler or hot water generator will be installed and tested without a long period of interim storage (see 3.18).
g. Fungus resistant varnish conforming to MIL-V-173 should be used to coat electrical components and circuit elements, including terminal and circuit connections, when the boiler is to be installed in humid conditions. Components and elements inherently inert to fungi or in hermetically sealed enclosures or current-carrying contact surfaces should not be coated.
h. Electromagnetic interference suppression (EMI), when required, should conform to the EMI suppression requirements and test limits for Class C3, Group I equipments as specified in MIL-STD-461. The boiler should be subjected to tests to determine conformance in accordance with MIL-STD-462.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, date, and number of this specification.
b. Types, sizes, and classes of steam boiler or hot water generator required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When the packaged unit is a steam boiler or hot water generator (see 3.1).
e. Thermal output capacity of steam boiler or hot water generator (see 3.1).
f. Fuel required (see 3.1).
g. When the steam boiler or hot water generator is to be designed for an indoor or outdoor installation (see 3.6).
h. Design parameters for steam boiler or hot water generator (see 3.6):

(1) Design pressure (psig).
(2) Operating pressure at packaged unit outlet (psig).
(3) Design temperature (^{o}F).
(4) Operating temperature at packaged unit outlet (^{o}F).
(5) Maximum continuous thermal output (Btu/hr).
(6) Feedwater temperature (^{o}F).
(7) Maximum, minimum, and average anticipated ambient air temperature (^{o}F).
(8) Site elevation above sea level (feet).

i. When flue gas stack exit temperatures are to be lower than as specified, with or without an economizer (see 3.7.4).
j. Emission requirements for opacity, particulates, nitrogen oxides, carbon monoxide, and sulfur dioxide (see 3.8).
k. When water pressure drop for hot water generator is to be other than specified (see 3.9.2).
l. Steam boiler or hot water generator trim and appurtances, if other than as specified (see 3.10).
m. When a manual feedwater regulator control is to be furnished (see 3.11.2).
n. Type of burner combustion control system required (see 3.12.1).
o. Location of control panel (see 3.12.2).
p. When mounted operating instructions are other than as specified (see 3.15).
q. When spare parts are required. If required, the type of spare parts and quantity to be provided (see 3.17).
r. When factory start up service is other than as specified (see 3.18).
s. Quality conformance inspection, if other than as specified (see 4.2.1).
t. On-site inspection, if other than as specified (see 4.2.2).
u. When all or part of the on-site inspection may be performed at the manufacturer's facilities (see 4.2.2).
v. When the examination of 4.3 and the test of 4.4 are to be repeated on-site (see 4.2.2).

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w. Testing procedures of the governing Federal, state, or local environmental agency for opacity, particulates, nitrogen oxides, carbon monoxide, and sulfur dioxide (see 4.5.8).
x. Level of preservation and level of packing required (see 5.1).
y. When receipt of the completed manufacturer's data reports for the specified steam boiler or hot water generator is required (see 6.3.1.2).

6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD Federal Acquisition Regulations (FAR) Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below should be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraph No.	Data Requirements Title	Applicable DID No.	Option
6.3.1.1, 6.3.1.2, 6.3.1.3 and 6.3.4	Certification Data/Report	DI-MISC-80678	None
6.3.3	Instruction, Equipment	UDI-E-21344 Installation	None
6.3.2	Commercial manuals	DI-M-2050	None

(DIDs related to this specification, and identified in section 6 will be approved and listed as such in DOD 5010.12-L, Vol. II, Acquisition Management Systems and Data Requirements Control List. Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3.1 Standards Compliance. Prior to approval of shipment, the contractor should submit for the approval of the contracting officer or authorized representative satisfactory evidence that the proposed steam boiler or hot water generator to be furnished under this specification meets the applicable requirements of UL-726, UL-795, Section I, Section IV, Section VIII, Division I, and Section IX of the ASME Code, and ASME CSD-1.

6.3.1.1 UL requirements. Acceptable evidence of meeting the requirements of UL-726 and UL-795 is the certification symbol, or listing mark on the products, or a certified test report from a recognized independent testing laboratory indicating that the steam boiler or hot water generator has been tested and conforms to UL requirements. Such evidence shall be acceptable to the contracting officer.

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6.3.1.2 ASME Code requirements. Acceptable evidence of meeting the requirements of Section I, Section IV, Section VIII, Division I, and Section IX of the ASME Code is the ASME code symbol and when specified (see 6.2), receipt of the completed manufacturer's data reports for the specified steam boiler or hot water generator.

6.3.1.3 ASME CSD-1 requirements. Acceptable evidence of meeting the requirements of ASME CSD-1 is the receipt of the manufacturer's certificate of compliance stating the equipment procured with this specification is in accordance with the applicable provisions of ASME CSD-1. Such evidence should be acceptable to the contracting officer.

6.3.2.1 Technical manuals. The manufacturer should provide three commercial technical manuals normally prepared and supplied with the steam boiler or hot water genertor.

6.3.3 Control panel skid-mounted junction box. Installation instructions covering the connection of the control panel should be provided when a free-standing remote mounted control panel is furnished with skid wiring or tubing terminates in a skid-mounted junction box (see 3.1.2.2).

6.3.4 Design verification. The manufacturer should furnish data to verify compliance with the specified heat input rate limit (see 3.7.3).

6.4 Boiler operational certification (Navy Only). The boiler operational certification should be performed, where applicable, per NAVFAC MO-324, after the boiler complies with the on-site tests (see 4.7). The boiler inspector should be certified by NAVFAC and licensed by the Engineering Field Division.

6.5 Definitions. Definitions used in this specification have been in accordance with the ABMA Lexicon of Boiler and Auxiliary Equipment. Other terms used in this specification should be interpreted in accordance with the referenced standards.

6.5.1 Effective heating surface. Effective heating surface is defined as the heat exchange surface directly exposed to the combustion gases on one side and to the water or water-steam mixture being heated on the other. Effective heating surface shall be calculated on the side receiving heat and shall neglect any extended surfaces.

6.6 Cross-reference of classifications. The classification of boilers in this document differs from the previous revision in the following respects:

MIL-B-17228E	MIL-B-17228F
TYPES	TYPES
Type I - Light-oil fired (Grade 2).	Type I - Steam boiler producing low pressure steam not greater than 15 psig.

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MIL-B-17228E (Continued)	MIL-B-17228F (Continued)
TYPES (Continued)	TYPES (Continued)
Type II - Gas-fired (natural, manufactured, mixed or liquefied petroleum gas as specified.	Type II - Steam boiler producing high pressure steam between 16 and 250 psig.
Type III Combination light-oil and gas-fired (grade 2 oil, type of gas), as specified.	Type III - Steam boiler producing high pressure steam between 251 and 350 psig.
Type IV - None	Type IV - Steam boiler producing high pressure steam between 351 and 600 psig.
Type V - None	Type V - Hot water generator producing low temperature water not greater than 160 psig or 250°F.
SIZES	SIZES
Size 1 - 400,000 to 2,500,000 Btu/hr	Size 1 - Thermal output capacity between 40,000 and 320,000 Btu/hr.
Size 2 - 2,500,001 to 5,000,000 Btu/hr	Size 2 - Thermal output capacity between 320,001 and 10,000,000 Btu/hr.
Size 3 - 5,000,001 to 8,000,000 Btu/hr	Size 3 - None
Size 4 - 8,000,0001 to 14,000,000 Btu/hr	Size 4 - None
Size 5 - 14,000,001 Btu/hr and above	Size 5 - None
CLASSES	CLASSES
Class 1 - Low pressure (15 psig operating pressure).	Class 1 - None
Class 2 - High pressure (150, 200, 250, or 300 psig design pressure.	Class 2 - Light oil fired.

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MIL-B-17228E (Continued)	MIL-B-17228F (Continued)
CLASSES (Continued)	CLASSES (Continued)
Class 3 - High-pressure hot water (160, 200, 250, 300, 400, or 500 psig design pressure as specified.	Class 3 - Gas fired.
Class 5 - None	Class 5 - Combination light oil and gas fired.

6.7 Subject term (key word) listing.

Boiler
Coiled tube
Gas and oil fired
High and low pressure
Hot water generator
Serpentine tube
Water tube

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - YD
Air Force - 99

User Activities:
Army - CE
Navy - MC

Preparing activity:

Navy - YD

(Project 4410-0074)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-B-17228F	2. DOCUMENT DATE (YYMMDD) 90/08/23

3. DOCUMENT TITLE Boilers, Steam and Hot Water, Watertube (Coil and Serpentine), Packaged Type (40,000 to 10,000,000 BTU/HR Thermal Output Capacity)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME R. B. REYNOLDS, Project Manager Facilities Criteria Development Division	b. TELEPHONE (Include Area Code) (1) Commercial (805) 982-5801 ext 428 (2) AUTOVON 551-5801 ext 428
c. ADDRESS (Include Zip Code) Commanding Officer (156) Naval Construction Battalion Center Port Hueneme, CA 93043-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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